TinyRF-KD: Protocol-Sensitive Knowledge Distillation of Ultra-Small Neural Detectors for Wireless Sensor Network Intrusion Detection with Dual-MCU Fixed-Point Validation

Nishant Harkut and Mahendra Shukla

***Abstract*—Wireless sensor network (WSN) intrusion detection under severe class imbalance demands detectors that fit kilobyte-scale memory and integer-arithmetic budgets while retaining accuracy on minority attack classes. This paper presents TinyRF-KD (*tiny RF knowledge distillation*), a controlled study of ultra-small multilayer-perceptron students distilled from a calibrated random-forest teacher on the WSN-DS corpus. Under train-only scaling with ten seeds, RF-KD Student A achieves macro-F1 0.9203 ± 0.0065 at $1{,}189$ parameters ($4.64\,$KB) and Student B achieves 0.9392 ± 0.0129 at $3{,}397$ parameters ($13.27\,$KB), representing $58.8\times$ and $20.6\times$ compression relative to the neural teacher.**

A critical finding is that KD benefit is protocol-contingent: seed-paired RF-KD minus scratch yields $\Delta = +0.0094$ ($p = 0.048$) under random-row splits but only $+0.0002$ ($p = 0.95$) under feature-group splits, and per-route RNG isolation independently confirms the collapse ($\Delta = +0.0007$ for A). Per-class analysis reveals minority-class tradeoffs: Student B RF-KD loses 0.023 Blackhole F1 relative to scratch while gaining Flooding and TDMA.

The fixed-point inference core cross-compiles for MSP430F1611-class motes in $2{,}842\,$B Flash. Full-test hardware-in-the-loop ($N = 56{,}200$) on ESP32-C3 and Arduino UNO R4 WiFi reproduces fixed-point references with agreement 1.0 on all four student–board pairs, with float-to-fixed macro-F1 reduction ≈ 0.024–0.027 and latencies from $116.5\,\mu$s to $791.4\,\mu$s.

Negative results are reported with the same rigor: global SHAP rank agreement is low and non-significant ($\rho \approx 0.24$, $p > 0.35$); co-distillation underperforms RF-KD by 0.0046–0.0069 macro-F1; curriculum-extension collapses at seed-5678 under multi-route training. Instrumented radio motes were unavailable, so commercial MCUs proxy the constrained on-node class via offline feature replay. These findings support KB-scale executable neural detectors when evaluation protocols, split constructions, and negative results are reported alongside compression ratios.

***Index Terms*—wireless sensor networks, intrusion detection, knowledge distillation, model compression, TinyML, fixed-point inference, hardware-in-the-loop, SHAP, Edge-IIoTset**

I. INTRODUCTION

THE proliferation of wireless sensor networks (WSNs) for environmental monitoring, industrial sensing, and infrastructure surveillance has elevated on-node security analysis to a first-order systems problem [1]. A practical detector must identify minority routing and traffic attacks from tabular features while its parameters, intermediate activations, and arithmetic fit devices with limited flash, RAM, and energy [2], [3]. On the public WSN-DS corpus [4], optimized ensembles and deep models report very high software accuracy [5]–[9]. These results establish a predictive ceiling but leave open a critical question: can a *kilobyte-scale integer student* preserve task utility under an auditable training protocol and full-test hardware replay?

Knowledge distillation transfers soft targets from a teacher to a compact student [10]. Stanton *et al.* demonstrated that student–teacher fidelity and generalization need not move together [11], framing KD as an empirical protocol question rather than a guaranteed improvement. Recent intrusion detection system (IDS) studies apply KD to reduce neural size on classical and IoT traffic benchmarks [12]–[17]. In parallel, hardware-aware IDS targets gateway-class platforms such as Raspberry Pi [18], hybrid designs retain deep models at a gateway after on-node filtering [19], and tree detectors have been embedded on ESP32-class devices for smart-home traffic [20].

What remains under-specified is the joint case that matters for WSN routing attacks: RF-to-tiny neural compression on WSN-DS, under a train-only scaler and leakage-aware protocol ladder, with two evaluation units for multi-seed tables versus deploy export, two-board full-test integer agreement, and a quantitative post-KD explanation-rank diagnostic. Explanation methods can disagree even when predictive metrics appear strong [21]–[23]. A responsible compression study must therefore report when distillation helps, when it does not, when integer conversion costs utility, and when global explanation ranks fail to transfer.

This paper studies that joint case under the name TinyRF-KD. Curriculum ordering and co-distillation appear as controlled routes, not as guaranteed superiority claims. Explainability enters as a measurement of global feature-rank transfer from RF TreeExplainer to student DeepExplainer, not as a claim that distillation preserves RF rankings. We do not claim first KD for IDS, first MCU IDS, SHAP rank preservation, co-distillation superiority under train-only scaling, or that deploy seed-42 macro-F1 is a multi-seed table cell.

The main contributions of this paper are as follows:

1) **Primary multi-seed compression (U-MS):** Fixed-capacity Students A/B under train-only scaler and ten seeds, with RF-KD as the principal distilled route, reported against scratch, MLP-KD, co-distillation, cur-

N. Harkut and M. Shukla are with the Atal Bihari Vajpayee Indian Institute of Information Technology and Management (ABV-IIITM), Gwalior 474015, India (e-mail: img_2023040@iiitm.ac.in, mahendra@iiitm.ac.in).

riculum, and control routes. Student A achieves $58.8\times$ compression (4.64 KB) with macro-F1 0.9203 ± 0.0065; Student B achieves $20.6\times$ compression (13.27 KB) with macro-F1 0.9392 ± 0.0129.

2) **Protocol sensitivity of KD benefit:** Seed-paired KD-minus-scratch contrasts random-row train-only against feature-group disjoint splits, revealing that KD gain for Student A is significant ($p = 0.048$) under random-row but vanishes ($p = 0.95$) under feature-group; per-route RNG isolation independently confirms the collapse ($\Delta = +0.0007$ for A, $+0.0005$ for B), with per-class costs showing minority-class tradeoffs under the stricter protocol.
3) **Dual evaluation protocols:** Multi-seed tables (U-MS) versus deploy protocol (U-DC) for hardware export, demonstrating that the deploy seed-42 macro-F1 0.9485 is not interchangeable with multi-seed results.
4) **Deployment feasibility and dual-MCU full-test HIL:** Fixed-point inference core cross-compiles for MSP430F1611-class WSN motes in 2,842 B Flash with $\sim$248 B stack, and full-test integer agreement on ESP32-C3 and Arduino UNO R4 WiFi ($N = 56{,}200$) is 1.0 on all four student–board pairs, with measured float-to-fixed utility cost 0.024–0.027 macro-F1 and board latencies from 116.5 μs to 791.4 μs.
5) **Negative and boundary results:** Global SHAP rank transfer is low and non-significant ($\rho \approx 0.24$, $p > 0.35$); co-distillation underperforms RF-KD by 0.0046–0.0069 macro-F1; curriculum-extension collapses at seed-5678 under multi-route training but recovers under isolated seeding; Edge-IIoTset group-aware evaluation shows lower absolute scores than standard random splits with identified $\approx 17\%$ test-row leakage.

The remainder of this paper is organized as follows. Section II reviews related work and positions our contributions. Section III defines the system model, evaluation boundary, and hardware constraints. Section IV presents the TinyRF-KD framework, training routes, and evaluation units. Section V details experimental protocols and dataset characteristics. Section VI reports comprehensive results including negative findings. Section VII discusses implications, limitations, and positioning against prior systems. Section VIII concludes with future directions.

II. Related Work

A. WSN-DS and Software Detectors

Almomani *et al.* introduced WSN-DS from NS-2/LEACH simulations with five traffic classes: Blackhole, Grayhole, Flooding, TDMA, and Normal [4]. Subsequent work emphasizes high software accuracy through resampling, feature selection, and ensemble methods [5]–[9], [24]–[26]. These papers establish a predictive ceiling against which ultra-small students should be judged but leave open whether kilobyte-scale neural students, under train-only scaling and two-board integer HIL, remain useful under explicit protocol controls. Our work uses WSN-DS as a comparable public benchmark and explicitly reports the train-only scaler and fixed split protocol.

B. Knowledge Distillation for IDS Compression

Hinton *et al.* formulated temperature-scaled soft-label distillation for model compression [10]. Stanton *et al.* showed that student–teacher fidelity and generalization need not move together, identifying optimization difficulties and dataset characteristics as key factors [11]. Yang *et al.* apply self-KD to lightweight network intrusion detection [12]. Wisanwanichthan and Thammawichai distill deep teachers into shallow networks across multiple IDS datasets on workstation hardware [13]. Yagiz and Goktas combine KD with variational autoencoders and attribution-based XAI [14]. Benaddi *et al.* couple SHAP-guided feature pruning with Kronecker-compressed students on TON_IoT [15]. Peng *et al.* study federated learning with KD under Non-IID IoT traffic [16]. Hossain *et al.* use SHAP-based knowledge sharing in federated botnet detection [17]. Guo *et al.* analyze why knowledge distillation works through attention and fidelity mechanisms [27]. TinyRF-KD differs by centering RF soft targets into fixed ultra-small MLPs on WSN-DS, a train-only multi-seed protocol with a feature-group stress test, two evaluation units, and two-MCU full-test integer HIL.

C. On-Device and Hardware-Aware IDS

Jacob *et al.* provide integer-arithmetic-only quantization foundations for neural network inference [28]. Diab *et al.* co-design tree and 1D-CNN detectors under flash/RAM/operation budgets and deploy on Raspberry Pi 3, achieving 95.3% accuracy with 75 KB flash and 1.2 K operations for LightGBM, and 97.2% accuracy with 190 KB flash and 840 K FLOPs for a HW-NAS-optimized CNN [18]. Alfarra and AbuSamra study energy-efficient hybrid WSN security with on-node rules and gateway CNN-LSTM [19]. Javed *et al.* embed CatBoost on an ESP32 smart thermostat for DoS/MITM detection on a home-IoT feature set, achieving 98.71% binary accuracy with 276 μs inference time [20]. Misrak *et al.* study quantization-aware hybrid IDS models with software size reports [29]. Banbury *et al.* and Sze *et al.* survey TinyML and efficient DNN processing constraints [2], [3]. Our HIL targets two MCU families as on-node proxies with full-test integer agreement for RF-distilled neural students, not gateway-class training/inference and not live radio energy accounting. We report inference latencies from 116.5 μs to 791.4 μs across four student–board pairs with full-test agreement 1.0.

D. Explainability and Disagreement

Lundberg and Lee introduced SHAP (SHapley Additive exPlanations) for model interpretation [23]. Krishna *et al.* document systematic disagreement among post-hoc explainers even when predictive metrics are strong [21]. Adebayo *et al.* provide sanity checks for saliency methods [22]. Nugraha *et al.* integrate SHAP for IDS feature workflows [30]. We measure global rank Spearman correlation between RF TreeExplainer and student DeepExplainer on the deploy unit and report low non-significant agreement ($\rho \approx 0.24$, $p > 0.35$) as a measured failure of global rank transfer, consistent with explanation disagreement literature.

III. System Model and Evaluation Boundary

A. Conceptual Placement

Fig. 1 distinguishes three layers of the problem. In a fielded WSN, sensing and short-range radio terminate at a sink or gateway, with optional cloud services beyond. An intrusion detector may run *on-node* (severe resource limits), at the *gateway* (SBC-class resources), or remotely. Hybrid designs intentionally leave deep models at the gateway [19]. Gateway-class hardware-aware search targets Pi-class platforms [18]. TinyRF-KD targets the on-node class: ultra-small students intended for constrained microcontrollers. The evaluation boundary is explicit: we measure model-core fixed-point fidelity and latency on commercial MCUs via offline feature replay, not over-the-air LEACH dynamics or joules-per-packet on a live mesh.

B. What We Instrumented

Instrumented radio motes with live mesh traffic were not available. We therefore evaluate inference fidelity on two commercial microcontrollers—ESP32-C3 and Arduino UNO R4 WiFi—using full-test replay of standardized WSN-DS feature vectors over USB serial. Fig. 2 shows the laboratory HIL bench. A Raspberry Pi host flashes firmware and streams vectors; it does *not* execute the student classifier for the reported MCU–reference agreement. That boundary is intentional so the HIL result remains interpretable: the Pi host flashes firmware and streams vectors, but the classifier runs only on the MCU under test.

C. Dataset and Threat Setting

WSN-DS provides a public five-class WSN routing/DoS setting with a large software literature, enabling protocol comparisons that a private mote capture would not support. Class counts in our repository CSV are 10,049 Blackhole, 3,312 Flooding, 14,596 Grayhole, 340,066 Normal, and 6,638 TDMA (374,661 rows, 17 numeric features after removing identifier and label). The Normal class dominates; minority attack classes drive macro-F1. We do not claim these rows are live over-the-air captures. Edge-IIoTset appears only as a secondary robustness study with its own feature construction and is never merged with WSN-DS scores.

IV. Proposed TinyRF-KD Framework

A. Problem Formulation

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$ denote the WSN-DS dataset with features $x_i \in \mathbb{R}^d$ ($d = 17$) and labels $y_i \in \{1, \ldots, C\}$ ($C = 5$ classes). Given a fixed stratified split into training $\mathcal{D}_{\text{train}}$, validation $\mathcal{D}_{\text{val}}$, and test $\mathcal{D}_{\text{test}}$ partitions, our objective is to train a compact neural network $f_\theta : \mathbb{R}^d \to \mathbb{R}^C$ parameterized by $\theta \in \mathbb{R}^{|\theta|}$ that minimizes the test macro-F1 while satisfying a memory constraint:

$$\min_{\theta} \; \mathcal{M}(f_\theta; \mathcal{D}_{\text{test}}) \quad \text{s.t.} \quad |\theta| \le \Theta_{\max}, \tag{1}$$

where $\mathcal{M}$ denotes macro-F1 and $\Theta_{\max}$ represents the parameter budget. We define the *compression ratio* relative to the neural teacher $f_{\theta_{\text{teacher}}}$ as:

$$\kappa = \frac{|\theta_{\text{teacher}}|}{|\theta_{\text{student}}|}. \tag{2}$$

B. Teachers and Students

The RF teacher uses 500 trees and maximum depth 15 [31]. For distillation, class probabilities are isotonic-calibrated with three-fold cross-validation [32]. A neural teacher MLP (17–128–256–128–5; 69,893 parameters; 273.02 KB FP32 parameter payload) provides an optional soft-target baseline. Student A is 17–32–16–5 (1,189 parameters, 1,136 MACs, 4.64 KB FP32). Student B is 17–64–32–5 (3,397 parameters, 3,296 MACs, 13.27 KB FP32). Architectures are fixed across routes; routes change losses and teachers, not capacity.

C. Training Routes and Losses

Let $z = f_\theta(x) \in \mathbb{R}^C$ denote the logits and $\sigma(z)_c = \exp(z_c)/\sum_{j=1}^{C} \exp(z_j)$ the softmax output. For knowledge distillation, we use temperature-scaled softmax:

$$\sigma^{(T)}(z)_c = \frac{\exp(z_c/T)}{\sum_{j=1}^{C} \exp(z_j/T)}, \tag{3}$$

where $T > 0$ controls the smoothness of the distribution. As $T \to \infty$, the distribution approaches uniform; as $T \to 0$, it approaches a one-hot distribution.

The scratch objective uses class-weighted cross-entropy:

$$\mathcal{L}_{\text{CE}}(\theta) = -\sum_{c=1}^{C} w_c \, y_c \log \sigma(f_\theta(x))_c, \tag{4}$$

where w_c are inverse-frequency class weights. For RF-KD, we combine hard and soft supervision as a convex balance:

$$\begin{aligned} \mathcal{L}_{\text{RF-KD}}(\theta) &= (1-\alpha)\, \mathcal{L}_{\text{CE}}(\theta) \\ &\quad + \alpha\, T^2 \, \mathrm{KL}\Big(q^{\text{RF}} \| \sigma^{(T)}\Big), \end{aligned} \tag{5}$$

where $\alpha \in [0, 1]$ balances hard and soft supervision. The temperature $T = 4$ and $\alpha = 0.7$ are used for all primary experiments, selected by grid search ($T \in \{2, 3, 4, 5\}$, $\alpha \in \{0.5, 0.7, 0.9\}$) against leakage-free validation. The KL divergence term encourages the student to match the teacher's predictive distribution:

$$\mathrm{KL}(p \| q) = \sum_{c=1}^{C} p_c \log \frac{p_c}{q_c}. \tag{6}$$

Co-distillation (route J) applies joint soft supervision from the RF teacher and a curriculum-trained MLP [15]:

$$\begin{aligned} \mathcal{L}_{\text{J}}(\theta) &= w_{\text{CE}}\, \mathcal{L}_{\text{CE}} + w_{\text{RF}}\, T^2 \mathrm{KL}(q^{\text{RF}} \| \sigma^{(T)}) \\ &\quad + w_{\text{CL}}\, T^2 \mathrm{KL}(q^{\text{CL}} \| \sigma^{(T)}), \end{aligned} \tag{7}$$

with $w_{\text{CE}} = 0.30$, $w_{\text{RF}} = 0.40$, $w_{\text{CL}} = 0.30$, where q^{CL} denotes soft targets from the curriculum-trained MLP.

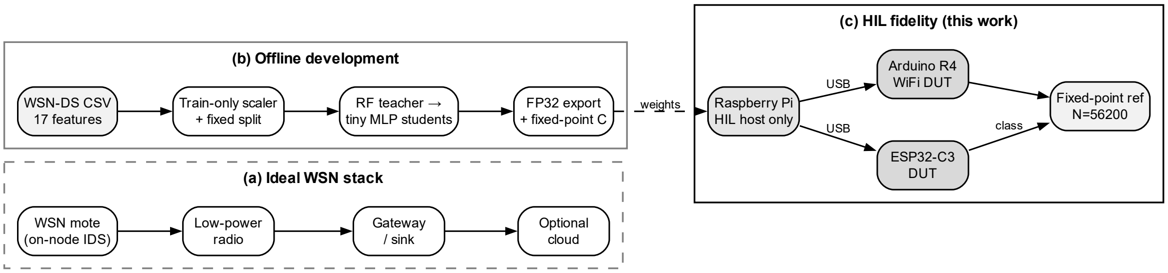

Fig. 1. System context for TinyRF-KD. (a) Ideal WSN stack with on-node detector placement. (b) Offline development on public WSN-DS features with train-only scaling and RF-to-MLP distillation. (c) HIL fidelity: Raspberry Pi host streams standardized test vectors over USB serial to ESP32-C3 and Arduino UNO R4 WiFi devices under test; predictions are compared to a host fixed-point reference ($N = 56{,}200$). The Pi is the orchestrator only, not the student classifier for agreement metrics.

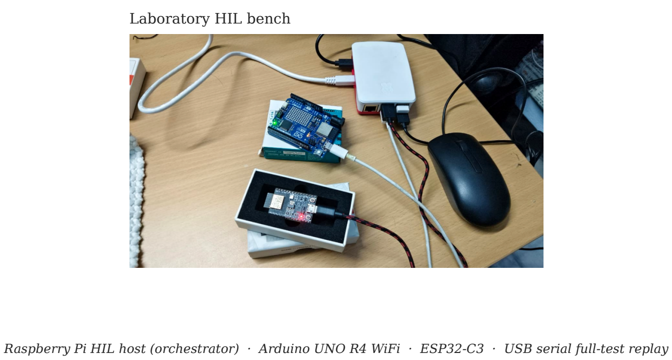

Fig. 2. Laboratory HIL bench used in this work: Raspberry Pi host with Arduino UNO R4 WiFi and ESP32-C3 devices under test (USB serial full-test replay of standardized feature vectors).

D. Fidelity and Agreement Metrics

Following Stanton *et al.* [11], we distinguish between *fidelity* (agreement with the teacher) and *generalization* (test performance). We define the agreement metric between a student f_θ and reference f_{ref} on test set $\mathcal{D}_{\text{test}}$:

$$\text{Agree}(f_\theta, f_{\text{ref}}) = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[\hat{y}_\theta(x_i) = \hat{y}_{\text{ref}}(x_i)], \tag{8}$$

where $N = |\mathcal{D}_{\text{test}}|$ and $\mathbf{1}[\cdot]$ is the indicator function. For MCU HIL, the reference f_{ref} is the host fixed-point model. We also report the predictive KL divergence:

$$\text{KL}_{\text{avg}}(f_\theta, f_{\text{ref}}) = \frac{1}{N}\sum_{i=1}^{N} \text{KL}(\sigma(f_{\text{ref}}(x_i)) \| \sigma(f_\theta(x_i))). \tag{9}$$

E. Train-Only Scaler

After the fixed seed-42 stratified 70/15/15 split, a `StandardScaler` is fit on the training partition only and applied to validation and test. Formally, the scaler computes mean μ_j and standard deviation σ_j for each feature j on $\mathcal{D}_{\text{train}}$:

$$\mu_j = \frac{1}{|\mathcal{D}_{\text{train}}|}\sum_{(x_i,y_i)\in\mathcal{D}_{\text{train}}} x_{i,j}, \tag{10}$$

$$\sigma_j = \sqrt{\frac{1}{|\mathcal{D}_{\text{train}}|}\sum_{(x_i,y_i)\in\mathcal{D}_{\text{train}}} (x_{i,j} - \mu_j)^2}, \tag{11}$$

and transforms all partitions: $\tilde{x}_{i,j} = (x_{i,j} - \mu_j)/\sigma_j$. This train-only protocol prevents test-set statistics from leaking into the model and is the default for all WSN-DS experiments.

F. Dual Evaluation Units

Fig. 3 defines two evaluation protocols on the same fixed stratified split:

- **Multi-seed protocol (U-MS):** Ten independent seeds with full route comparison (scratch, MLP-KD, RF-KD, co-distillation, curriculum). Each seed trains all routes before moving to the next.
- **Deploy protocol (U-DC):** Single seed-42 RF-KD only, used for ONNX export, fixed-point conversion, and two-board HIL.

The distinction is critical: seed-42 multi-seed RF-KD macro-F1 0.9249 is *not* the deploy protocol seed-42 macro-F1 0.9485.

G. Export, Fixed-Point, and HIL Protocol

deploy FP32 graphs are exported for host ONNX Runtime and OpenVINO evaluation. MCU export uses post-training fixed-point quantization. Given FP32 weights $w \in \mathbb{R}$ and activations $a \in \mathbb{R}$, we map to fixed-point:

$$w_{\text{fix}} = \text{round}(w/s_w), \tag{12}$$

$$a_{\text{fix}} = \text{clip}\big(\text{round}(a/s_a), -2^{n-1}, 2^{n-1} - 1\big), \tag{13}$$

where $s_w, s_a > 0$ are scale factors determined by calibration, and $n = 8$ for Q8. The scale factor is computed from calibration data $\mathcal{D}_{\text{cal}}$:

$$s = \frac{\max_{x\in\mathcal{D}_{\text{cal}}} |f(x)|}{2^{n-1} - 1}, \tag{14}$$

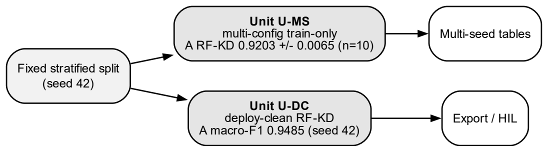

Fig. 3. Two evaluation protocols on the same fixed stratified split. Multi-seed (U-MS) supports distributional analysis; deploy (U-DC) supports hardware deployment. Student A RF-KD scores differ between protocols and must not be interchanged.

ensuring the dynamic range covers observed activation magnitudes. We disclose a float-to-fixed macro-F1 drop bound of 0.03; a stricter 0.01 bound is not claimed for these weights.

Quantization-aware training probes reduced deploy FP32 macro-F1 to 0.8996 (Student A, $\Delta = -0.018$) and 0.9213 (Student B, $\Delta = -0.023$), worse than post-training quantization at 0.9244/0.9180; these QAT weights were not used for the reported HIL binaries.

Before MCU flashing, the fixed-point C kernel is cross-compiled for an MSP430F1611-class target (TelosB/Tmote Sky-style WSN mote, 48 KB Flash, 10 KB RAM) to verify memory feasibility on genuinely constrained hardware. Under `-Os` with `msp430-elf-gcc` 9.3.1, the linked smoke firmware (integer StandardScaler preprocessing plus fixed-point MLP inference) requires 2,842 B Flash-class `.text`, 0 B `.data`, 6 B `.bss`, and a compiler-reported bounded project-function stack usage of approximately 248 B during prediction. This cross-compile result supports MSP430/TelosB-class memory feasibility of the model core; it does not constitute physical deployment or measure real-time latency and energy on a live mote.

For each student–board pair, the host streams all 56,200 test vectors; the MCU returns predictions; agreement is scored against the generated fixed-point reference. Reported boards: ESP32-C3 and Arduino UNO R4 WiFi. Algorithm 1 summarizes the end-to-end process.

V. Experimental Protocols

Table I summarizes evaluation contracts. Ten seeds for primary WSN-DS multi-seed work:

$$\mathcal{S} = \{42, 123, 456, 789, 1001, 2024, 3141, 5678, 8192, 9999\}.$$

Feature-group disjoint sensitivity uses five seeds $\{42, 123, 456, 789, 1001\}$ and 56,301 test rows. Edge-IIoTset appears only as a secondary robustness study with its own feature construction and 15-class label space.

Primary metric is macro-F1 (equal class weight), motivated by the severe class imbalance in WSN-DS (Normal: 91% of rows). Macro-F1 is defined as the harmonic mean of per-class precision and recall, averaged across classes:

$$\text{macro-F1} = \frac{1}{C}\sum_{c=1}^{C} \frac{2 \cdot \mathrm{Prec}_c \cdot \mathrm{Rec}_c}{\mathrm{Prec}_c + \mathrm{Rec}_c}, \tag{15}$$

Algorithm 1 TinyRF-KD training and dual-MCU HIL workflow

Require: WSN-DS features, fixed seed-42 split, student capacity $\{A, B\}$, seed set $\mathcal{S}$
Ensure: Multi-seed tables (U-MS), deploy artifacts (U-DC), HIL agreement matrix

```
Fit StandardScaler on training partition only; transform val/test
for each seed s ∈ S under multi-seed protocol do
    Train RF teacher; calibrate soft targets; train routes (scratch, MLP-KD, RF-KD, J, curriculum, controls)
    Record accuracy, macro-F1, per-class F1
end for
Aggregate U-MS means and sample standard deviations; compute seed-paired KD−scratch tests
deploy unit: set_seed(42); train RF-KD only for A and B
Export FP32; evaluate ONNX/OpenVINO; convert to fixed-point C under 0.03 macro-F1 drop bound
for each board ∈ {ESP32-C3, Arduino R4} and student ∈ {A, B} do
    Flash firmware; stream N = 56,200 test vectors over USB serial
    Score MCU predictions vs fixed-point reference; record latency
end for
On U-DC, compute global SHAP rank Spearman (RF TreeExplainer vs student DeepExplainer)
Secondary: Edge-IIoTset group-aware splits and leakage audit (no merge with WSN-DS scores)
return tables, figures, HIL matrix, SHAP diagnostics
```

where Prec_c and Rec_c are precision and recall for class c. Accuracy is reported for continuity with prior work. Per-class F1 exposes minority costs that macro-F1 can hide. Multi-seed summaries use mean $\pm$ sample standard deviation.

For seed-paired comparisons, we use the paired t-test:

$$t = \frac{\bar{\Delta}}{s_\Delta/\sqrt{n}}, \quad \bar{\Delta} = \frac{1}{n}\sum_{i=1}^{n}(\mathrm{F1}^{(i)}_{\mathrm{KD}} - \mathrm{F1}^{(i)}_{\mathrm{scratch}}), \tag{16}$$

where n is the number of seeds and s_Δ is the sample standard deviation of the paired differences. We also report two-tailed Wilcoxon signed-rank tests as a non-parametric alternative. Given shared splits and modest n, p-values are interpreted as exploratory rather than confirmatory.

VI. Results and Analysis

This section follows the evaluation contracts of Section V. We report primary multi-seed compression, protocol sensitivity and minority costs, full multi-seed route surface, two-unit and seeding diagnostics, host conversion, two-MCU HIL, SHAP rank transfer, and Edge-IIoTset secondary study. Positive compression and HIL fidelity results are reported alongside negative or boundary findings with the same evidence level.

TABLE I
EXPERIMENTAL PROTOCOLS

Protocol	Rows	Classes	Features	Seeds	Notes
WSN-DS multi-seed (U-MS)	374,661	5	17	10	Train-only scaler; fixed stratified 70/15/15.
WSN-DS feature-group	374,661	5	17	5	Disjoint on identical feature vectors.
WSN-DS deploy (U-DC)*	374,661	5	17	1	Seed-42 RF-KD only; ONNX/HIL.
Edge-IIoTset group-aware	2,219,201	15	49	5	Removes cross-partition groups.

*Same fixed stratified split as U-MS; differ only in seed count and route set.

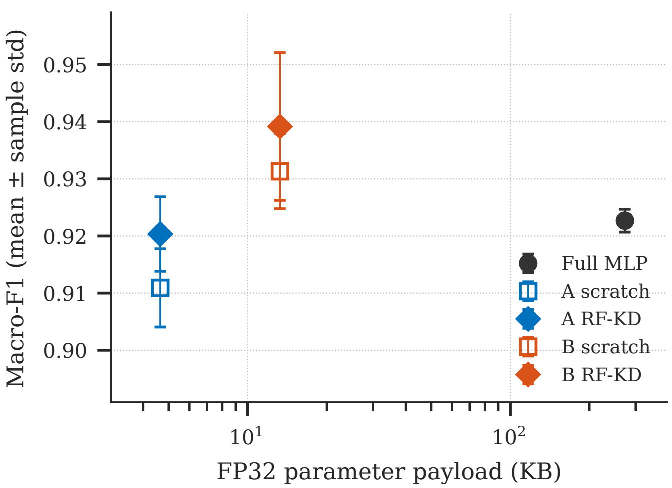

Fig. 4. Size–performance under U-MS (train-only multi-seed). Points from aggregate CSVs; open markers: scratch; filled: RF-KD. RF teacher omitted because its size basis is serialized pickle, not FP32 parameter payload.

A. Primary Multi-Seed Compression

Table II and Fig. 4 summarize the primary train-only multi-seed unit (U-MS). The RF teacher remains strongest (0.9789 ± 0.0003 macro-F1). The full MLP reaches 0.9227 ± 0.0020. Student A RF-KD reaches 0.9203 ± 0.0065 macro-F1 at 4.64 KB FP32 parameters, above Student A scratch (0.9109 ± 0.0068). Student B RF-KD reaches 0.9392 ± 0.0129, above Student B scratch (0.9313 ± 0.0066). Relative to the full MLP FP32 payload, Students A and B achieve $58.8\times$ and $20.6\times$ compression.

Fig. 5 compares the principal student routes. Co-distillation (J) yields Student A macro-F1 0.9158 ± 0.0026 and Student B 0.9323 ± 0.0141, underperforming RF-KD by 0.0046 (A) and 0.0069 (B) in mean macro-F1. We therefore do not claim co-distillation superiority under train-only scaling.

B. Protocol Sensitivity of KD Benefit

Table III and Fig. 6 report seed-paired RF-KD minus scratch. Under random-row splits, Student A paired mean $\Delta = +0.00944$ (t-test $p = 0.0479$, Wilcoxon $p = 0.0488$, $n = 10$). Under feature-group splits, paired mean $\Delta = +0.00023$ (t-test $p = 0.946$, Wilcoxon $p = 1.0$, $n = 5$). Student B RF-KD minus scratch is $+0.00783$ (t-test $p = 0.097$, $n = 10$) under random-row and -0.00165 (t-test $p = 0.735$, $n = 5$) under feature-group. KD benefit is therefore split-dependent: significant under random-row but vanishing under feature-group.

C. Per-Class Costs under Feature-Group Partitions

Fig. 7 shows RF-KD minus scratch per class under feature-group disjoint five-seed evaluation. Student B RF-KD loses 0.0232 Blackhole F1 and 0.0158 Grayhole F1 relative to scratch while gaining Flooding (+0.0100) and TDMA (+0.0201). Student A shows a larger Blackhole loss (−0.0362) with gains on Flooding and TDMA. Macro-F1 alone can hide minority degradation; protocol-sensitive KD analysis must therefore report class-level costs.

D. Complete Route Tables

Tables IV and V list all multi-seed routes, including curriculum and control routes. Curriculum-extension rows exhibit large standard deviations driven by the seed-5678 collapse discussed in Section VI-E. These full tables make the multi-seed surface auditable rather than limited to the selected main routes in Table II.

E. Curriculum-Extension Instability

In the multi-seed protocol, curriculum-extension at seed 5678 collapses to macro-F1 ≈ 0.4075 with Blackhole and Grayhole F1 equal to zero. Isolated re-runs with seed-5678 recover healthy macro-F1 ≈ 0.9145 across five trials. We attribute the collapse to multi-route training-path contingency, not to inherent seed failure under single-route training. Curriculum-extension is reported for integrity, not as a recommended default.

Isolated RF-KD training for Student A averages macro-F1 0.9138 ± 0.0056, versus multi-seed RF-KD 0.9203 ± 0.0065 (paired $\Delta = -0.0066$, t-test $p = 0.037$). For seed 42 alone, isolated RF-KD is 0.9148 while the multi-seed cell is 0.9249 and deploy protocol is 0.9485. These gaps justify dual-protocol reporting.

Critically, the corresponding isolated scratch (per-route seed) baseline is 0.9131 ± 0.0055 for Student A and 0.9307 ± 0.0046 for Student B, yielding an isolated KD-minus-scratch Δ of only $+0.0007$ (A) and $+0.0005$ (B)—essentially zero. This provides a second, independent confirmation that the KD advantage vanishes: under per-route RNG isolation (a completely different correction mechanism than feature-group splitting), the RF-KD gain collapses to near-zero, consistent with the feature-group disjoint result ($\Delta = +0.0002$, $p = 0.95$).

TABLE II
SELECTED WSN-DS TEN-SEED RESULTS UNDER TRAIN-ONLY SCALER (U-MS). NEURAL SIZES ARE FP32 PARAMETER PAYLOADS. RF SIZE IS MEAN SERIALIZED PICKLE (DIFFERENT BASIS).

Model	Route	Params	Size (KB)	Accuracy	Macro-F1
RF teacher	supervised	–	85186[†]	0.9966 ± 0.0001	0.9789 ± 0.0003
Full MLP	supervised	69,893	273.02	0.9873 ± 0.0003	0.9227 ± 0.0020
Student A	scratch	1,189	4.64	0.9844 ± 0.0024	0.9109 ± 0.0068
Student A	MLP-KD	1,189	4.64	0.9854 ± 0.0003	0.9120 ± 0.0023
Student A	RF-KD	1,189	4.64	0.9869 ± 0.0010	0.9203 ± 0.0065
Student A	co-distill (J)	1,189	4.64	0.9861 ± 0.0004	0.9158 ± 0.0026
Student B	scratch	3,397	13.27	0.9887 ± 0.0010	0.9313 ± 0.0066
Student B	MLP-KD	3,397	13.27	0.9864 ± 0.0007	0.9177 ± 0.0046
Student B	RF-KD	3,397	13.27	0.9902 ± 0.0020	0.9392 ± 0.0129
Student B	co-distill (J)	3,397	13.27	0.9889 ± 0.0023	0.9323 ± 0.0141

[†]Serialized RF artifact; not comparable to raw neural FP32 payloads on the same axis.

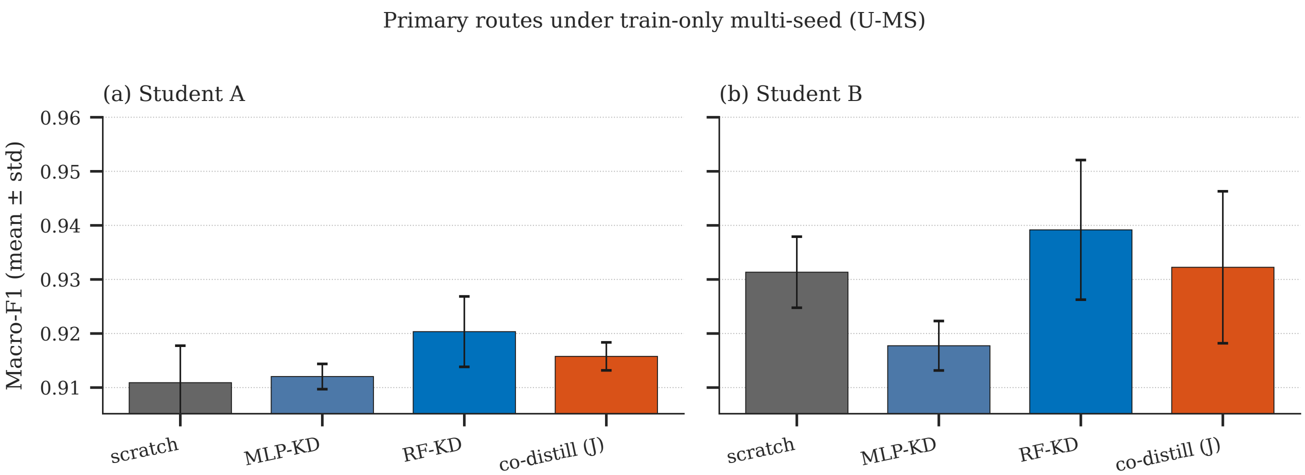

Fig. 5. Primary student routes under U-MS. Co-distillation (J) underperforms RF-KD for both students under the train-only multi-seed protocol.

TABLE III
SEED-PAIRED RF-KD MINUS SCRATCH MACRO-F1

Student	Split protocol	Δ mean	t p	n
A	random-row	$+0.00944$	0.048	10
A	feature-group	$+0.00023$	0.946	5
B	random-row	$+0.00783$	0.097	10
B	feature-group	-0.00165	0.735	5

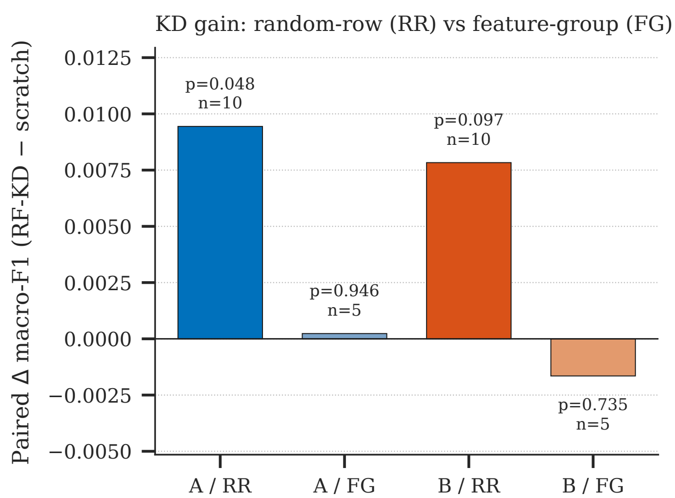

Fig. 6. Paired Δ macro-F1 (RF-KD − scratch) under random-row versus feature-group splits. KD benefit for Student A is significant under random-row and vanishes under feature-group.

F. Deploy Protocol Host Conversion

On U-DC, PyTorch FP32 macro-F1 is 0.9485 (A) and 0.9449 (B) with $N = 56{,}200$. Fig. 8 summarizes host conversion. ONNX Runtime FP32 matches PyTorch predictions with agreement 1.0 for both students at serialized model sizes of 5.44 KB (A) and 14.07 KB (B), with median inference latency 27.0 μs (A) and 21.4 μs (B) on the host workstation. Dynamic INT8 ONNX reduces macro-F1 to 0.8938 (A) and 0.9066 (B) and is retained only as a software-size probe. OpenVINO FP32 agreement with PyTorch is 1.0 in the deploy protocol.

G. Dual-MCU Full-Test HIL

Table VI and Fig. 9 report the four-pair matrix under the deploy protocol. MCU versus fixed-point reference agreement is 1.0 for every pair; all statuses OK; $n = 56{,}200$. MCU fixed-point macro-F1 is 0.9244 (A) and 0.9180 (B) on both boards (board-invariant reference match). Mean total latency

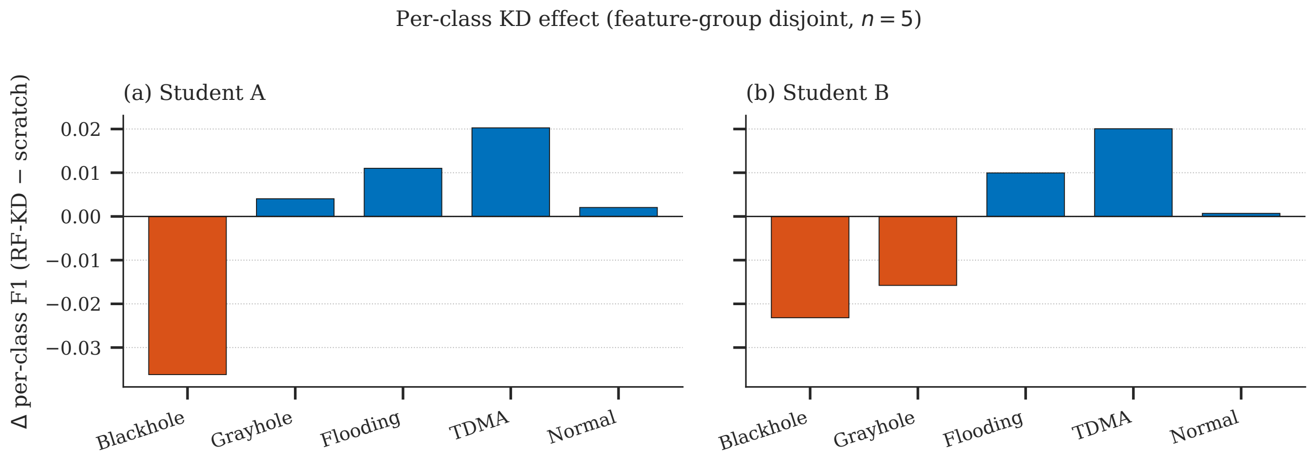

Fig. 7. Per-class Δ F1 (RF-KD − scratch) under feature-group disjoint partitions ($n = 5$). Minority Blackhole/Grayhole losses can offset Flooding/TDMA gains.

TABLE IV
ALL STUDENT A ROUTES UNDER TRAIN-ONLY MULTI-SEED AGGREGATE (U-MS)

Config	Params	Size (KB)	Accuracy	Macro-F1
A_RF_500	–	85186	0.9966 ± 0.0001	0.9789 ± 0.0003
B_Full_MLP	69893	273.02	0.9873 ± 0.0003	0.9227 ± 0.0020
C2_CL_MLP_domain	69893	273.02	0.9857 ± 0.0007	0.9131 ± 0.0045
C_CL_MLP_loss	69893	273.02	0.9857 ± 0.0007	0.9136 ± 0.0039
C_CL_MLP_loss_ext	69893	273.02	0.9786 ± 0.0252	0.8674 ± 0.1617
C_CL_MLP_loss_fair	69893	273.02	0.9857 ± 0.0007	0.9136 ± 0.0039
D_Small_MLP (scratch)	1189	4.64	0.9844 ± 0.0025	0.9109 ± 0.0068
E2_KD_from_MLP	1189	4.64	0.9854 ± 0.0004	0.9120 ± 0.0023
E_KD_from_RF	1189	4.64	0.9869 ± 0.0011	0.9203 ± 0.0065
F_KD_from_CL_MLP	1189	4.64	0.9851 ± 0.0004	0.9105 ± 0.0031
F_KD_from_CL_MLP_ext	1189	4.64	0.9812 ± 0.0127	0.8937 ± 0.0554
F_KD_from_CL_MLP_fair	1189	4.64	0.9851 ± 0.0004	0.9105 ± 0.0031
G_KD_random_pacing	1189	4.64	0.9852 ± 0.0006	0.9108 ± 0.0038
I_KD_from_SMOTE_MLP	1189	4.64	0.9847 ± 0.0008	0.9085 ± 0.0050

TABLE V
ALL STUDENT B ROUTES UNDER TRAIN-ONLY MULTI-SEED AGGREGATE (U-MS)

Config	Params	Size (KB)	Accuracy	Macro-F1
D_Small_MLP (scratch)	3397	13.27	0.9887 ± 0.0010	0.9313 ± 0.0066
E2_KD_from_MLP	3397	13.27	0.9864 ± 0.0008	0.9177 ± 0.0046
E_KD_from_RF	3397	13.27	0.9902 ± 0.0021	0.9392 ± 0.0129
F_KD_from_CL_MLP	3397	13.27	0.9862 ± 0.0008	0.9166 ± 0.0051
F_KD_from_CL_MLP_ext	3397	13.27	0.9831 ± 0.0105	0.9045 ± 0.0421
F_KD_from_CL_MLP_fair	3397	13.27	0.9862 ± 0.0008	0.9166 ± 0.0051
G_KD_random_pacing	3397	13.27	0.9861 ± 0.0008	0.9166 ± 0.0049
I_KD_from_SMOTE_MLP	3397	13.27	0.9864 ± 0.0014	0.9176 ± 0.0088

is $116.5\,\mu$s (ESP32-C3 A), $320.3\,\mu$s (ESP32-C3 B), $301.5\,\mu$s (R4 A), and $791.4\,\mu$s (R4 B). Fixed-versus-FP32 prediction agreement on export vectors is 0.9919 (A) and 0.9905 (B). Float-to-fixed macro-F1 reduction relative to deploy FP32 is approximately 0.024 (A: $0.9485 \rightarrow 0.9244$) and 0.027 (B: $0.9449 \rightarrow 0.9180$), within the disclosed 0.03 bound and outside a 0.01 bound. Input quantization uses signed Q8 calibrated standardized features with zero reported saturation on 955,400 values in the export reports. MCU HIL agreement to the fixed reference remains 1.0, isolating remaining utility loss to conversion rather than board execution. Compiled firmware footprints on the target boards are 281,776–284,132 B Flash (21% of 1,310,720 B) and 13,592 B RAM (4% of 327,680 B) on ESP32-C3, and 56,384–58,736 B Flash (21%–22% of 262,144 B) and 7,128 B RAM (21% of 32,768 B) on Arduino UNO R4 WiFi, confirming that the full HIL firmware including serial protocol and model weights occupies well under half of each board's available memory.

H. SHAP Rank Transfer

SHAP (SHapley Additive exPlanations) assigns each feature an importance value based on Shapley values from cooperative game theory [23]. For a model f and input x, the SHAP value $\phi_j(f, x)$ for feature j is:

$$\phi_j(f,x) = \sum_{S \subseteq [d] \setminus \{j\}} \frac{|S|!\,(d - |S| - 1)!}{d!} \big[f(x_{S \cup \{j\}}) - f(x_S)\big], \tag{17}$$

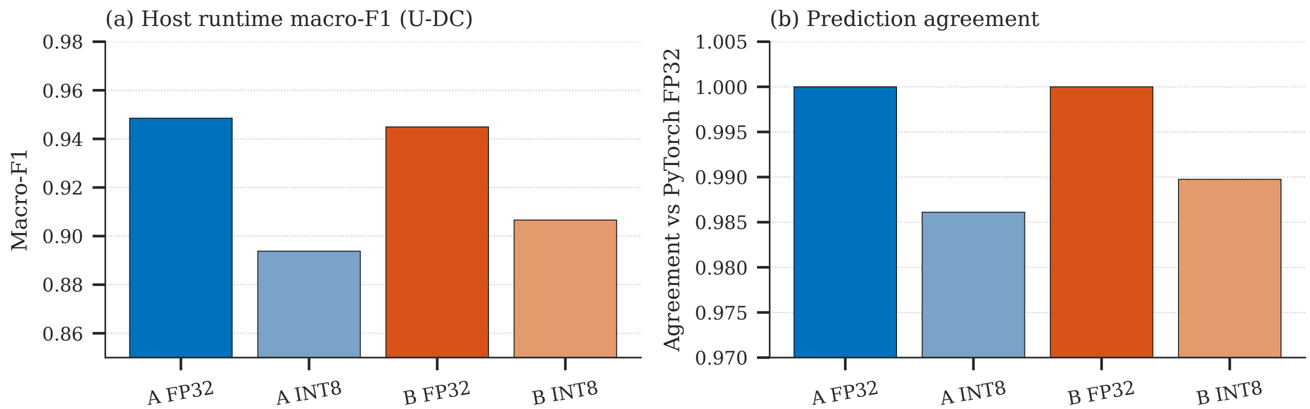

Fig. 8. Host conversion on U-DC RF-KD graphs (ONNX Runtime). FP32 matches PyTorch at agreement 1.0; dynamic INT8 reduces macro-F1 substantially and is not used for MCU HIL binaries.

TABLE VI
FULL-TEST HIL UNDER DEPLOY TRAIN-ONLY RF-KD ($N = 56{,}200$)

Board	Student	Agree (MCU/ref)	Acc.	Macro-F1	Mean lat. (μs)
ESP32-C3	A	1.0	0.9875	0.9244	116.5
ESP32-C3	B	1.0	0.9873	0.9180	320.3
Arduino R4	A	1.0	0.9875	0.9244	301.5
Arduino R4	B	1.0	0.9873	0.9180	791.4

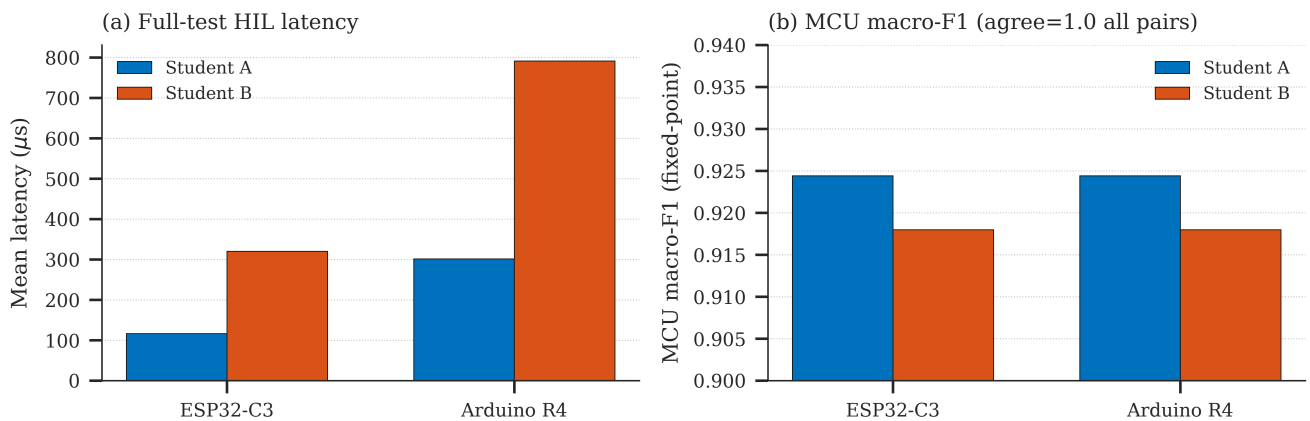

Fig. 9. Dual-MCU HIL under U-DC RF-KD ($N = 56{,}200$). (a) Mean latency. (b) MCU fixed-point macro-F1 (agreement with fixed reference is 1.0 on all four pairs).

where x_S denotes the input with features not in S replaced by background values and $[d] = \{1, \ldots, d\}$. For tree models, TreeExplainer computes exact Shapley values in polynomial time. For neural networks, DeepExplainer approximates Shapley values using backpropagation.

We measure global rank agreement using Spearman correlation on mean absolute SHAP values:

$$\rho = \frac{\sum_{j=1}^{d}(R_j^{\mathrm{RF}} - \bar{R}^{\mathrm{RF}})(R_j^{\mathrm{student}} - \bar{R}^{\mathrm{student}})}{\sqrt{\sum_j (R_j^{\mathrm{RF}} - \bar{R}^{\mathrm{RF}})^2}\sqrt{\sum_j (R_j^{\mathrm{student}} - \bar{R}^{\mathrm{student}})^2}}, \tag{18}$$

where R_j is the rank of feature j by mean $|\phi_j|$ across the test set.

Fig. 10 and Table VII summarize the explanation-rank diagnostic. Global mean-absolute SHAP rank Spearman correlation between RF TreeExplainer and student DeepExplainer on U-DC is $\rho = 0.238$ ($p = 0.358$) for Student A and $\rho = 0.225$ ($p = 0.384$) for Student B. Bootstrap means over five repeats are 0.178 ± 0.086 (A) and 0.162 ± 0.055 (B). Top features diverge: Student A global top features include `Is_CH`, `JOIN_S`, `dist_CH_To_BS`; the RF teacher top features include `Is_CH`, `ADV_S`, `Data_Sent_To_BS`, `DATA_S`, `Rank`. Shared presence of `Is_CH` does not imply global rank agreement. We treat this as a measured failure of global rank transfer, consistent with explanation disagreement literature [21], not as evidence that RF-KD preserves RF explanations.

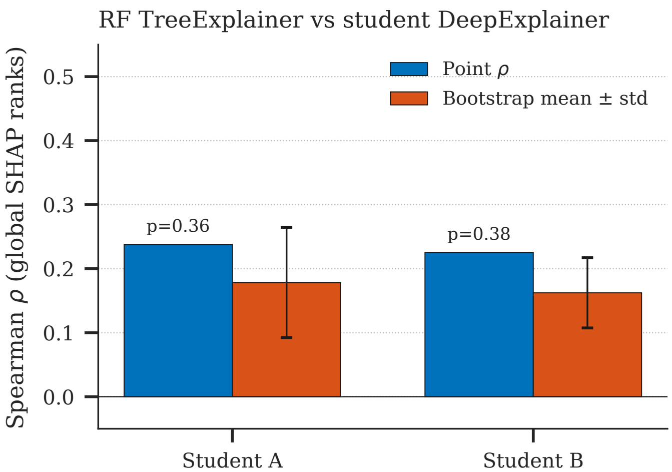

Fig. 10. Global SHAP rank agreement on U-DC (RF TreeExplainer vs student DeepExplainer). Point Spearman ρ is low and non-significant; bootstrap means confirm weak transfer.

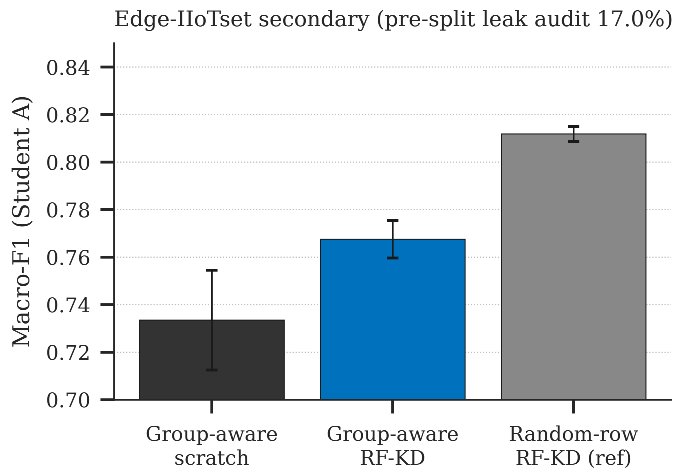

Fig. 11. Edge-IIoTset Student A: group-aware versus standard random splits. Group-aware evaluation removes $\approx 17\%$ test-row leakage.

TABLE VII
TOP-5 GLOBAL FEATURES BY MEAN |SHAP| ON DEPLOY RF-KD (U-DC)

Rank	Student A	RF teacher
1	`Is_CH`	`Is_CH`
2	`JOIN_S`	`ADV_S`
3	`dist_CH_To_BS`	`Data_Sent_To_BS`
4	`Data_Sent_To_BS`	`DATA_S`
5	`ADV_S`	`Rank`

I. Edge-IIoTset Secondary Study

Under standard random splits, pre-encode cross-partition groups expose about 17.0% of test rows. Group-aware splits remove that exposure (0% leakage). Fig. 11 shows Student A results. Student A RF-KD mean macro-F1 under group-aware five seeds is 0.7676 ± 0.0079 with KD-minus-scratch mean $+0.0340$; standard random-split RF-KD for Student A is 0.8118 ± 0.0032. Student B under the same group-aware protocol achieves 0.8129 ± 0.0068 RF-KD macro-F1 with KD-minus-scratch mean only $+0.0073$, substantially smaller than Student A's gain and consistent with the capacity-contingent KD benefit observed on WSN-DS. Note: the group-aware RF teacher used 200 trees versus 500 in the reference, so the comparison involves two configuration changes. Absolute scores are lower under group-aware evaluation; KD still shows a positive mean delta for Student A but a marginal one for Student B. Edge-IIoTset scores are never merged with WSN-DS scores.

J. Deploy Protocol Replication

With matching soft targets, ten-seed RF-KD means are 0.9206 ± 0.0127 (A) and 0.9370 ± 0.0126 (B), with seed-42 exactly reproducing 0.9485 (A) and 0.9449 (B). This confirms that the high deploy seed-42 point is reproducible under matching soft targets.

VII. DISCUSSION

A. What the Evidence Supports

Ultra-small RF-KD students retain competitive macro-F1 under train-only multi-seed evaluation at 4.64 KB and 13.27 KB FP32 payloads, achieving $58.8\times$ and $20.6\times$ compression relative to the neural teacher while preserving detection performance. Dual-board full-test integer agreement is achievable for the deploy RF-KD unit, with float-to-fixed macro-F1 cost 0.024–0.027 inside a disclosed 0.03 bound and inference latencies from 116.5 μs to 791.4 μs. Host FP32 conversion preserves predictions at agreement 1.0. These results support KB-scale executable neural detectors when evaluation units and protocol controls are reported carefully.

B. What Requires Qualification

KD gain vanishes under two independent corrections: feature-group splits ($\Delta = +0.0002$, $p = 0.95$) and per-route RNG isolation ($\Delta = +0.0007$ for A, $+0.0005$ for B). Either correction alone would be noteworthy; both converging on near-zero strengthens the conclusion that the multi-seed random-row KD advantage is protocol-contingent. KD benefit can also hide minority-class costs (Blackhole/Grayhole losses under feature-group). Deploy seed-42 macro-F1 (0.9485 for Student A) is not a multi-seed table cell (0.9203 ± 0.0065 under U-MS). Global SHAP ranks are not preserved ($\rho \approx 0.24$, $p > 0.35$). Co-distillation (J) underperforms RF-KD ($\Delta = -0.0046$ for A, -0.0069 for B). Curriculum-extension can collapse under multi-route training (seed-5678 $\rightarrow$ 0.41 macro-F1) even when isolated re-seeding recovers (≈ 0.91).

C. Relation to Closest Prior Systems

Javed *et al.* demonstrate tree IDS on ESP32 for a different dataset and task, achieving 98.71% binary accuracy with 276 μs inference [20]. Wisanwanichthan and Thammawichai demonstrate KD compression in software on workstation hardware [13]. Diab *et al.* demonstrate hardware-aware models on Pi-class gateways with 95.3% accuracy at 75 KB flash and

1.2 K operations for LightGBM [18]. TinyRF-KD sits between these lines: RF-distilled neural students, WSN routing-attack labels, protocol ladder, two evaluation units, two-MCU HIL, and an explicit offline-feature evaluation boundary. We do not claim first MCU IDS, first KD for IDS, or gateway-class deployment.

D. Limitations

Instrumented WSN motes and live radio energy measurements were unavailable; commercial MCUs proxy the constrained on-node class via offline feature replay. Shared fixed splits limit claims of independence across seeds. SHAP uses finite background and explain sample sizes and focuses on global ranks rather than local explanations. HIL latencies are board means under serial replay, not worst-case real-time radio stacks. Edge-IIoTset results use a different feature space and label set than WSN-DS and are not merged.

VIII. Conclusion and Future Work

TinyRF-KD provides a protocol-honest account of RF-to-tiny neural distillation for WSN-DS intrusion detection. Under multi-seed evaluation, Students A and B retain strong macro-F1 at $4.64\,\mathrm{KB}$ and $13.27\,\mathrm{KB}$ FP32 payloads ($58.8\times$ and $20.6\times$ compression), but KD benefit is protocol-contingent: significant under random-row ($p = 0.048$ for A) but vanishing under both feature-group splits ($\Delta = +0.0002$, $p = 0.95$) and per-route RNG isolation ($\Delta = +0.0007$ for A, $+0.0005$ for B)—two independent corrections that converge on the same conclusion. A deploy protocol supports MSP430F1611-class cross-compile in $2{,}842\,\mathrm{B}$ Flash and dual-MCU full-test integer agreement (1.0 on all four student–board pairs) with measured float-to-fixed utility cost 0.024–0.027 macro-F1. Global SHAP ranks do not transfer reliably ($\rho \approx 0.24$, $p > 0.35$). Co-distillation does not dominate RF-KD, and curriculum-extension instability under multi-route training must be disclosed. These findings support KB-scale executable detectors when evaluation protocols, splits, and negative results are reported with the same rigor as compression ratios.

Future work includes instrumented radio motes with live mesh traffic and energy accounting, on-device feature extraction from packet streams, broader multi-dataset evaluation, and tighter quantization-aware training under the same two-protocol reporting.

Data and Artifact Availability

Primary multi-seed aggregates, split comparisons, per-class deltas, deploy artifacts, HIL summaries, Edge results, SHAP data, and figure provenance are provided in the accompanying project repository at https://github.com/nishantharkut/CuKD-XAI.

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